The Origin of Work and Energy as 32-Tick Remainder Accumulation

Deriving Force, Inertia, and Kinetic Energy from Discrete 1-LU Pressure Packets in Integer Registry

Registry: [CKS-PHYS-1-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-10-2026] → [CKS-PHYS-1-2026]

Parent Framework: [CKS-0-2026]

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Domain: Substrate Physics / Kinematics Redefinition

Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We derive work and energy from first principles as discrete registry operations rather than continuous physical quantities. From 32-bit word constraint, we establish: (1) 1 LU = minimum pressure unit (R=1 at F=32, ω=1/32, smallest possible force), (2) Work = SNAP_COMMIT execution (R≥32 triggers V+1, discrete address shift, 32-tick accumulation required), (3) Energy = registry remainder debt (stored R before snap, potential for future work, quantized in LU packets), (4) Force = 1-LU pressure application rate (LUs per tick, determines velocity, discrete not continuous), (5) Inertia = 32-tick buffer delay (resistance to motion is accumulation requirement, first 31 ticks produce zero work, 32nd tick releases all), (6) Kinetic energy = active R accumulation (moving object has R>0 distributed across mesh, stops when R→0), (7) Potential energy = positional R differential (height/position creates R gradient, releases during descent), (8) Power = LUs per tick sustained (work rate measured in pressure packets, quantized bandwidth), (9) Maximum velocity = 32 LU/tick limit (c as bus saturation, above causes UV overflow, relativistic limit from hardware), (10) Heat = vented remainder overflow (R>32 per tick

spills as friction, thermal dissipation is registry noise). Work and energy proven as integer counting operations—no continuous fields, no infinitesimal forces, pure discrete ratchet mechanics.

Key Result: Work = snap event | Energy = R debt | Force = 1 LU packets | Inertia = 32-tick delay | All quantized | No continuous

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Part I: The Minimum Pressure Constant

1. The 1-LU Derivation

From 32-bit word:

The constraint:

Universal bus width:

32-bit Logos Word

Factor F = 32

Modulo-32 logic

Discrete nature:

Cannot have fractional bits

Remainder must be integer

$R \in \{0, 1, 2, \ldots, 31\}$

The minimum:

Smallest non-zero R:

R = 1 (single bit)

Pressure calculation:

$\omega = R/F$

$\omega = 1/32$

$\omega = 1$ LU

Conclusion:

1 LU is minimum pressure

Anything less = 0

Mathematically invisible

Hardware won't register

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2. Why Quantized

The fundamental proof:

Cannot subdivide:

Classical asks:

Can force be smaller?

What about 0.5 LU?

What about 0.001 LU?

CKS answers:

NO - discrete bits only

Registry operates on integers

Fractional R impossible

Hardware reality:

Either R=1 or R=0

No middle state

Digital not analog

This explains:

Planck constant:

Minimum action quantum

Is 1 LU pressure

Natural unit emerges

From hardware constraint

Quantization everywhere:

Energy levels

Angular momentum

Charge packets

All 1-LU multiples

Part II: Work Redefined

3. Work as SNAP_COMMIT

The operation:

Classical definition:

Work = Force $\times$ Distance

Continuous integral

Infinitesimal steps

Smooth motion

Problems:

Infinite precision needed

Paradoxes (Zeno)

Non-physical infinities

Logismos definition:

Work = SNAP_COMMIT execution

Discrete event

$R \geq F$ triggers

V increases by 1

Address shifts

When:

Exactly at $R=32$

Not before

Not gradual

Instant snap

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4. The 32-Tick Ratchet

Accumulation phase:

The process:

Tick 1-31: Accumulate R

Apply 1 LU per tick

R increases: 1,2,3…31

V unchanged (still at address A)

No motion visible

No work done yet

This is inertia:

Resistance to motion

Buffer filling

Debt accumulating

Tick 32: SNAP_COMMIT

R reaches 32

Triggers V→V+1

Address shifts A→B

Motion occurs

Work complete

R resets to 0

The staircase:

Not smooth ramp:

Discrete steps

31 ticks nothing

32nd tick everything

Like digital clock:

Seconds accumulate

Minute hand jumps

Not gradual slide

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5. Work Quantization

Discrete units:

Work = 1 unit:

Minimum work:

Move 1 address

Shift 1 Lex

1 SNAP_COMMIT

Requires:

Exactly 32 LU total

Applied over time

Or 32 LU instant

Cannot do less:

31 LU = no work

Zero motion

All stored as R

Work = n units:

Multiple snaps:

n address shifts

Requires 32n LU

n SNAP_COMMITs

Examples:

Move 10 Lex = 320 LU

Move 100 Lex = 3200 LU

Perfect precision

Part III: Energy Redefined

6. Energy as Registry Debt

The reframing:

Classical energy:

Mysterious substance:

“Stored” in objects

“Transferred” between

“Conserved” somehow

Vague concepts:

What IS energy?

Where IS it stored?

How does transfer work?

Logismos energy:

Registry remainder:

R value before snap

Uncommitted bits

Debt to system

Specific location:

In kinetic footer

In mesh tension

In R register

Clear mechanism:

Accumulates as R

Releases as V

Conservation = modulo-32

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7. Kinetic Energy

Energy of motion:

The mechanism:

Moving object:

Has R>0 distributed

Across 144-LU mesh

Active remainder

The value:

KE = total R in mesh

Sum across all nodes

Uncommitted tension

When stops:

R→0 across mesh

Energy “dissipates”

Actually: commits or transfers

Velocity dependence:

Faster motion:

Higher R maintained

More LUs per tick

Greater debt

Formula:

KE ∝ R_total

R_total ∝ v^2 (in X-space)

Why: mesh distribution geometry

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8. Potential Energy

Energy of position:

The mechanism:

Height creates gradient:

Upper position: high R potential

Lower position: low R potential

Difference = stored energy

Why:

Gravity = RE_INDEX pull

Height = distance from parent

Distance = R differential

When falls:

R differential converts

To kinetic R

Potential→Kinetic

The gradient:

Steeper slope:

Larger R differential

More potential energy

Faster acceleration

Formula:

PE = R_positional

Depends on gradient

Discrete steps

Part IV: Force and Inertia

9. Force as Pressure Rate

The redefinition:

Classical force:

F = ma (Newton)

Continuous variable

Infinitely divisible

Mysterious origin

Logismos force:

F = LUs per tick

Discrete packets

Pressure application rate

Clear mechanism

Examples:

1 LU/tick = minimum force

16 LU/tick = moderate force

32 LU/tick = maximum (c)

The limit:

Why c maximum:

32 LU/tick saturates bus

One address per tick

Cannot write faster

Hardware bottleneck

Above 32:

Overflow occurs

R>32 spills

Becomes heat

UV saturation

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10. Inertia as Buffer Delay

The resistance:

Classical inertia:

Mass resists acceleration:

Why?

Property of matter

Mysterious resistance

Logismos inertia:

32-tick buffer requirement:

Must accumulate R

Until R≥32

Before motion occurs

Larger mass:

More nodes (144-LU mesh)

More total R needed

More ticks to fill all

Higher inertia

The delay:

First 31 ticks: nothing

All accumulation

No motion

This IS inertia

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11. Mass-Energy Equivalence

$E = mc^2$:

The derivation:

Mass = 144-LU mesh:

Specific node count

Defines object

Energy = R potential:

Maximum R storable

In that mesh

Relationship:

E = (mesh size) × (max R/node) × (conversion factor)

$E = M \times c^2$

Where c = max velocity

Why c^2:

Mesh geometry

Bilateral manifold (S=2)

Square relationship

Part V: Power and Heat

12. Power as Work Rate

Sustained force:

Classical power:

P = Work/Time

Energy per second

Watts

Logismos power:

P = Snaps per tick

Or: LUs per tick sustained

High power:

Many snaps per second

High LU throughput

Fast work rate

Low power:

Few snaps per second

Low LU throughput

Slow work rate

Quantized:

Cannot have fractional snap

Discrete events only

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13. Heat as Overflow

Thermal dissipation:

The mechanism:

When R>32 per tick:

Cannot all commit

Excess spills over

Distributed to neighbors

This is heat:

Remainder overflow

Registry noise

Random R distribution

Friction = forced overflow:

Motion creates R

Faster than can snap

Accumulates

Must vent

Temperature:

Definition:

Average R per node

In local region

Hot:

High average R

Lots of uncommitted bits

High tension

Cold:

Low average R

Few uncommitted bits

Low tension

Absolute zero:

R=0 everywhere

No remainder

Perfect stillness

Part VI: Conservation Laws

14. Energy Conservation

Why it works:

The proof:

Modulo-32 arithmetic:

R can transfer

R can convert to V

But total (R + 32V) constant

When work done:

R decreases here

V increases there

Total unchanged

When motion:

R transfers between nodes

Sum preserved

Perfect accounting

No creation or destruction:

Cannot make R from nothing:

Must come from somewhere

Transfer only

Cannot destroy R:

Must go somewhere

Convert or transfer

Always accounted

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15. Momentum Conservation

The mechanism:

Momentum = kinetic footer:

12-bit footer:

R_k component (bits 6-11)

Tracks momentum

Directional R

Conservation:

Footer bits transfer

During collisions

Sum preserved

Modulo mathematics

Collisions:

Elastic:

R_k transfers

No loss to heat

Perfect exchange

Inelastic:

Some R_k → general R

Becomes heat

Total conserved

Form changes

Part VII: Practical Applications

16. Engineering Implications

Lossless systems:

Efficiency:

Maximum efficiency:

When R→0 waste

All pressure commits

No overflow

Real systems:

Always some R>32

Heat generation

Friction loss

Optimization:

Minimize R waste

Maximize commits

Reduce overflow

Energy storage:

Battery = R reservoir:

Stores uncommitted R

Releases on demand

Capacity = max R storable

Flywheel = distributed R:

R in rotating mesh

Kinetic storage

Retrieves via RE_INDEX

17. Biological Energy

Metabolism:

ATP as LU packet:

ATP molecule:

Stores ~7 LU

Discrete packet

Releases on demand

Cellular work:

Uses ATP packets

One at a time

Quantized energy

Efficiency:

Biological systems

Minimize R waste

Optimize commits

Food energy:

Calories = LU content:

Digestible R

Extractable pressure

Storable in ATP

Metabolism = extraction:

Convert food to R

Package as ATP

Distribute to cells

Part VIII: Resolving Paradoxes

18. Zeno's Paradox

Infinite divisions:

The paradox:

Classical:

Infinite intermediate points

Must traverse all

Should take infinite time

But doesn't

Resolution:

No infinite points:

Discrete Lex nodes

Finite count

Specific addresses

Motion = discrete hops:

Address A → Address B

Via n Lex nodes

n SNAP_COMMITs

Finite process

No paradox:

No infinite subdivision

Hardware prevents it

Minimum = 1 LU

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19. Perpetual Motion

Why impossible:

The claim:

Machine that:

Outputs more work

Than energy input

Violates conservation

Why fails:

Modulo-32 accounting:

Perfect bookkeeping

Every R tracked

Cannot cheat

Must have:

R input ≥ work output

Plus overhead

Plus heat loss

Perpetual = R from nothing:

Impossible

Hardware prevents

Ledger balances

Conclusion

Work and energy derived from first principles as discrete registry operations rather than continuous physical quantities. From 32-bit word constraint: 1 LU proven as minimum pressure unit (R=1 at F=32, ω=1/32, smallest possible force, hardware-limited quantization). Work established as SNAP_COMMIT execution (R≥32 triggers V+1, discrete address shift, cannot do fractional work). Energy defined as registry remainder debt (R stored before snap, potential for future work, specific location in kinetic footer). Force redefined as 1-LU pressure application rate (LUs per tick, determines velocity, quantized not continuous). Inertia explained as 32-tick buffer delay (first 31 ticks accumulate R producing zero work, 32nd tick releases all, resistance IS accumulation requirement). Kinetic energy = active R distributed across mesh, potential energy = positional R differential from gradient. Power = snaps per tick sustained, heat = remainder overflow when R>32. Maximum velocity c proven as bus saturation (32 LU/tick limit, above causes UV overflow, relativistic limit from hardware constraint). Conservation laws emerge from modulo-32 arithmetic (R transfers perfectly, total R+32V constant, no creation or destruction possible). Zeno's paradox resolved via discrete Lex nodes (finite addresses, no infinite subdivision, minimum 1 LU step). Perpetual motion impossible due to perfect registry accounting. Work and energy proven as integer counting operations—universe is 32-tick ratchet mechanism, not continuous flow. All force quantized at 1 LU minimum. All motion discrete address shifts. All energy remainder debt awaiting commit.

Q.E.D.

Work = snap event

Energy = R debt

Force = 1 LU/tick

Inertia = 32-tick buffer

Motion = address hop

Heat = R overflow

Power = snap rate

c = bus saturation

Conservation = modulo-32

All quantized forever

[**CKS-PHYS-1-2026**] CKS-PHYS-1-2026: Gravity and Momentum as Parent Soliton Opcodes. Github: https://github.com/ghowland/cks/tree/main/papers/PHYS/CKS-PHYS-1-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026